
Algorithm 2 Multi-Checkpoint Fingerprint Collection

Input: context c , token t , model θ , checkpoints $C = \{c_1 < \dots < c_m = L\}$, projection dim $k \ll D$, projection seed σ'

- 1: $a^{(1)}, \dots, a^{(L)} \leftarrow \text{ForwardPass}(\theta, c, t)$
 - 2: **for** $c_i \in C$ **do**
 - 3: $f_i \leftarrow P_{\sigma'+c_i} a^{(c_i)}$
 - 4: **end for**
 - 5: **return** $f = (f_1, \dots, f_m) \in \mathbb{R}^{mk}$
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$$\mathbb{E}_\pi[\text{verifier cost}] = \sum_i \pi_i K(c_i) \ll K(L)$$

Algorithm 3 Randomized-Depth Fingerprint Matching

Input: context c , token t , model θ , checkpoints C , projection dim k , seed σ' , audit distribution π over C , reference fingerprints $f \in \mathbb{R}^{mk}$

- 1: $d \sim \pi$
 - 2: $\hat{a}^{(1)}, \dots, \hat{a}^{(d)} \leftarrow \text{PartialForwardPass}(\theta, c, t, d)$
 - 3: **for** $c_i \in C$ with $c_i \leq d$ **do**
 - 4: $\hat{f}_i \leftarrow P_{\sigma'+c_i} \hat{a}^{(c_i)}$
 - 5: $d_i \leftarrow \|f_i - \hat{f}_i\|_2$
 - 6: **end for**
 - 7: **return** $\{d_i : c_i \leq d\}$
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$$\bar{\Pi}_i := \Pr_{\pi}[d \geq c_i], \quad \Pr[\text{escape}] = (1 - \bar{\Pi}_i)^N$$